MILITARY SPECIFICATION

CLOTH, KNIT, JERSEY, ARAMID

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an aramid, jersey, knit cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-831 - Test Report, Preparation of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

FEDERAL REGULATIONS

Rules and Regulations under the Textile Fiber Products Act.

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580).

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample of the jersey knit cloth shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Material. The cloth shall be produced from spun yarn of 1.5 denier per filament, two inch cut length, high temperature resistant meta-aramid fiber with a minimum twist of 28 turns per inch. The yarn shall not char at temperatures below 357°C (675°F).

3.3 Color. Unless otherwise specified, the color of the cloth shall be natural (undyed).

3.4 Physical properties. The physical properties shall conform to table I, when tested as specified in 4.6.

3.5 Width. Unless otherwise specified, the overall minimum width shall be 50 ± 2 inches (see 6.2.1).

3.6 Finish. The cloth shall be scoured, desized, dipped in a melamine stabilizing resin and heat set to meet the requirements for dimensional stability (see Table I). The weight of the stabilizing melamine resin shall be approximately 6% of the fabric weight.

3.7 Length and put up. Unless otherwise specified, the cloth shall be furnished in continuous lengths, each not less than 50 yards, and shall be put up in rolls as specified in PPP-P-1133.

3.8 Fiber identification. Each roll shall be labeled for fiber content in accordance with the Textile Fiber Products Identification Act.

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workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Test equipment and inspection facilities. The manfacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. The first article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirement of this specification (see 4.4).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on the individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all testing shall be performed under standard atmospheric conditions and performed on specimens in moisture equilibrium under standard atmospheric conditions specified in FED-STD-191.

4.4 First article inspection. First article inspection of the aramid, knit, jersey cloth shall consist of examinations and tests for all of the requirements of this specification.

4.4.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit three linear yards of the knit cloth. The sample shall be representative of the construction, workmanship, components and materials to be used during production. When a contractor is in continuous production of this knit cloth from contract to contract, submission of further first article inspection samples, on the new contract, may be waived at the discretion of the acquiring activity (see 6.2.1). Approval of the first article inspection samples or the waiving of the first article inspection does not waive the requirements for performing the quality conformance inspection. The first article inspection

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sample shall be furnished to the Government as directed by the contracting officer (see 6.2.1). A test report, containing complete test data for each component required herein, shall be prepared in accordance with MIL-STD-831. The test report shall accompany the first article samples.

4.4.2 Disposition of first article samples. Upon completion of the first article inspection recommendations and comments pertinent for use in monitoring production will be forwarded by the Government activity responsible for the inspection program (see 6.2.1) to the contracting officer. The sample will be consumed or destroyed during testing and shall not be considered as part of the quantity to be delivered under the contract or order.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations in 4.5.1 through 4.5.1.4 and all the tests required for the cloth for each lot. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.5.1 Examination of the end item. Examination of the end item shall be in accordance with 4.5.1.1 through 4.5.1.4.

4.5.1.1. Yard by yard examination. The required yardage of each roll shall be examined on the face side and visual defects as defined herein shall be classified as listed in Table II. All defects found shall be counted regardless of their proximity one to another except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with Inspection Level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 2.5 major defects and 6.5 total defects per 100 units (yards). The total size shall be expressed in units of one linear yard. An approximate equal number of yards shall be examined from each roll selected for yard by yard examination.

4.5.1.2 Overall examination. Each defect listed in Table IV shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size and the acceptance number shall be specified in Table III.

4.5.1.3 Examination for length.

4.5.1.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The maximum number of defects acceptable in the sample shall be in accordance with Table III.

4.5.1.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths in rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

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4.5.1.4 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this Act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.5.2 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of PPP-P-1133.

4.6 Test methods. The methods of testing specified in FED-STD-191, as listed in Table V, shall be followed. The physical and chemical values specified in Table I, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3-1/2 yards (full width) of the finished cloth and 1/4 yard (full width) of unfinished cloth. The lot size shall be expressed in units of one yard. The lot shall be unacceptable if one or more units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sampling size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

5. PACKAGING

5.1 Preservation, packing and marking. Cloth shall be preserved, packed and marked in accordance with PPP-P-1133. The levels of preservation and packing shall be as specified by the contracting activity (see 6.2.1).

6. NOTES

6.1 Intended use. The cloth is intended for use as the backing for the laminated cloth used in the fabrication of the continuous-wear anti-exposure coveralls.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity and width.
c. Selection of applicable levels of preservation and packing (see 5.1).
d. Whether first article is waived (see 4.4.1).
e. Name and address of the first article inspection laboratory (see 4.4.1); including the Government Activity responsible for conducting the inspection program (see 4.4.2).

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.2.2 Data requirements. When this specification is used in an sition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in acccordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement	Applicable DID no.
4.4.1	First article inspection reports	DI-T-5329

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of one sample unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Knitting information. This cloth has been successfully knit on a 28 cut circular, single, jersey knit machine using 50^s single cotton count yarn spun from 1.5 denier per filament, two inch cut length, PT-450 Nomex ® aramid fiber from E.I. Dupont DeNemours Company.

Preparing activity:
Navy - AS
(Project 8305-N996)

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TABLE I. Physical properties

Characteristics	Requirements
Weight, ounces per square yard	1.7 ± .20
Wales per inch	30 ± 2
Courses per inch	33 ± 2
Flame resistance, seconds, maximum:	
After flame time	2.0
After glow time	25.0
Char length, inches, maximum	5.0
Type of knit	Jersey
Dimensional stability, percent, maxiumum: 1/	
Wales direction	6.0
Courses direction	1.5
Nonfibrous material, percent, maximum	1.0 2/
Curling	Shall lie flat without distortion and show no evidence of curling.
pH	5.0 - 8.0 3/

1/ The preshrinkage process shall not be identified by name or trademark either on the cloth or on the ticket.

2/ Including starch and protein matter as well as chloroform and water soluble material.

3/ Determined on the water extract of the cloth.

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TABLE II. Classification of defects.

Defect	Classification	
	Major	Minor
Any hole, cut, or tear	X	
Run or dropped stitch, or end out	X	
Tucking defect		X
Float, pulled stitch or needle streak		X
Mend	X	
Abrasion 1/	X	
Thin place 1/	X	
Thin yarn 1/	X	
Tight yarn 1/	X	
Any slub or slug, more than twice the size of normal yarn		X
Foreign matter 1/		X
Knots extending above the surface of the cloth 1/		X
Kinks or untrimmed ends		X
Crease or wrinkle (embedded)		X
Stop mark		X
Tension too tight or too loose		X
Barre or shade bar 1/		X
Spot or stain	X	
Bowing		X

1/ Clearly visable at normal inspection distance (3 feet).

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TABLE III. Sample size.

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If lot is fewer than three rolls, each roll in the lot shall be examined.

TABLE IV. Overall defects.

Objectionable odor
Overall uncleanness
Uneven knitting throughout the piece
Off shade, shaded end to end, side to side, side to center or throughout the roll.
Poor dye penetration, mottled, streaky or cloudy
Overall width less than specified

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Table V. Test methods.

Characteristic	FED-STD-191 Test Method
Denier per filament	1/
Twist	1/
Cut length of filament	1/
Material identification	1/
Weight	5041
Wales	5070
Courses	5070
Flame resistance	5903
Nonfibrous material	2611
Dimensional stability	5556 2/
Curl	3/
pH	2811

1/ Unless otherwise specified, a certificate of compliance may be submitted for the stated requirement.

2/ 15 cotton laundering procedures shall be followed.

3/ Two specimens of cloth, 1-1/2 inches wide by 6 inches long shall be cut, one having the long dimension parallel to the wales and the other with the long dimension perpendicular to the wales. Both shall be placed on a flat surface for at least five minutes and then visually examined for evidence of curling.

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